

VLE and flash calculations

After midterm · Session 1 of 6 · 180 minutes

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Session outcome

Predict a two-phase state, solve it, then check both component balances.

Before class: Review chemical potential, fugacity and activity standards in Modules 2–3.

Learning path and all labs

The 180-minute class

IN CLASS	MINUTES
Recall and prediction	10
Concepts and derivation	45
Worked example	30
Break	10
Instructor lab demonstration	25
Guided student exploration	35
Discussion and interpretation	15
Exit question and independent task	10

The equilibrium problem

At fixed temperature and pressure, a stable equilibrium minimizes total Gibbs energy subject to material conservation.

$$\mu_i^L = \mu_i^V \quad \Longleftrightarrow \quad \hat{f}_i^L = \hat{f}_i^V$$

A calculation also needs a phase model, composition constraints and a stability check.

Liquid and vapor descriptions

$$y_i \hat{\phi}_i P = x_i \gamma_i P_i^{sat} \phi_i^{sat} \Pi_i$$

- Ideal vapor, ideal liquid: all correction factors are one.
- Modified Raoult: retain liquid activity coefficients.
- Gamma-phi: also retain vapor and reference-state corrections.

Lab 01 uses ideal vapor. Lab 09 adds second-virial vapor corrections.

Four boundary calculations

SPECIFIED	UNKNOWN BOUNDARY	COMPOSITION SUPPLIED
T	Bubble P	Liquid x
T	Dew P	Vapor y
P	Bubble T	Liquid x
P	Dew T	Vapor y

A dew calculation generally needs an inner liquid-composition iteration because γ depends on x .

Ideal-mixture boundary equations

$$P_{bubble} = \sum_i x_i P_i^{sat}$$

$$\frac{1}{P_{dew}} = \sum_i \frac{y_i}{P_i^{sat}}$$

Both equations are for fixed T, ideal liquid and ideal vapor. Using the same numerical composition in each equation describes different incipient-phase states.

TP flash includes a material balance

$$z_i = (1 - \beta)x_i + \beta y_i, \quad y_i = K_i x_i$$

$$\sum_i \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0$$

Here β is vapor mole fraction. In a nonideal liquid, K depends on the unknown x , so composition and phase fraction must be solved together.

A reproducible ideal baseline

Lab 01 synthetic binary at 350 K: $P_1^{sat} = 150$ kPa and $P_2^{sat} = 60$ kPa. Choose ideal liquid, $z_1 = 0.5$, $P = 95$ kPa.

QUANTITY	EXPECTED RESULT
Bubble P at $x_1=0.5$	105 kPa
Dew P at $y_1=0.5$	85.7143 kPa
Flash x_1, y_1	0.388889, 0.614035
Vapor fraction β	0.493506

The values are synthetic definitions, not real-fluid measurements.

A check independent of the solver

$$0.5 = (1 - 0.493506)(0.388889) + (0.493506)(0.614035)$$

The rounded result closes the component-1 balance. Component 2 must close as well.

Also check $x_1 + x_2 = y_1 + y_2 = 1$ and equality of component fugacities. A small balance residual alone does not validate the physical model.

Activity models change the liquid

One-parameter Margules, two-parameter Margules and NRTL share the same equilibrium criterion but predict different $\gamma(x)$.

Hold the synthetic component properties and T fixed. Change only the liquid model or one interaction parameter.

A nonconvex liquid needs an LLE/VLLE stability analysis; it cannot simply be accepted as a stable VLE flash.

Vapor corrections need a reference

$$\ln \hat{\phi}_i = \frac{P}{RT} \left(2 \sum_j y_j B_{ij} - B_{mix} \right)$$

$$B_{mix} = \sum_i \sum_j y_i y_j B_{ij}$$

Lab 09 also evaluates pure saturated-vapor ϕ consistently. With all $B_{ij}=0$, the ideal-vapor limit must be recovered.

A model comparison has controlled inputs

Compare the same T, P, feed and liquid model. Change only the vapor description first.

Record differences in boundary pressures, phase compositions and β . Then examine the declared B(T) range and the truncation guard.

PR/SRK, dense vapor and virial adiabatic flash are outside these lab solvers.

Class demonstration • Labs 01 and 09

[Open Lab 01](#) • [Open Lab 09](#)

1. Reproduce the ideal baseline and save its results.
2. Cross a phase boundary by changing pressure.
3. In Lab 09, first use zero B_{ij} , then change only B_{12} .

Before each run, predict the direction of change.

Guided exploration • 35 minutes

In pairs, choose one baseline state and one controlled change.

- One student predicts; the other records units and model settings.
- Recalculate and independently check a component balance.
- Exchange roles and explain any unexpected result.

Use the guided worksheet in the lab and save the calculator export separately.

Adiabatic flash as an extension

$$H_{feed} = (1 - \beta)H^L(T, x) + \beta H^V(T, y)$$

At specified pressure and feed enthalpy, T is another unknown. A TP flash at the feed temperature does not impose this energy balance.

Lab 07 uses synthetic caloric data. Independent practice only if the core TP calculation is secure.

Exit question

Two calculations have the same T and P . One returns a bubble-point vapor composition; the other returns a flash vapor fraction.

What additional information distinguishes these problems?

What evidence would make a converged flash result unsuitable for use?

Independent practice • suggested 60–90 minutes

Assessment set 1 and rubric • Offline package

One baseline and one changed vapor model; report phase, x , y , vapor fraction and balance residual.

Retain the calculator export, your worksheet, a comparison plot/table and one independent check.

State an assumption that limits your conclusion.

Use the core labs on the [learning path](#). Optional extensions are additional work.

References and further study

[Module reference deck](#) · [Lab sources and equations](#)

Derivations and original figure references remain in the corresponding module deck. Each lab records its implemented equations and assumptions.

Synthetic worked examples illustrate calculations; they are not evidence of real-system accuracy.